



CORBA Assignment — Step-by-Step Execution Guide

I read your uploaded files:

- AddImpl.java
- Client.java
- Server.java

Your assignment is a:

CORBA Addition Calculator

where:

- Client sends two numbers
- Server adds them remotely
- Result returns to client



First Understand Your Project Structure

Your project should contain:

```
DS_CORBA
├── src
│   ├── Add.idl
│   ├── AddImpl.java
│   ├── Client.java
│   ├── Server.java
│   └── AddModule
```



COMPLETE EXECUTION GUIDE

We will use:

Component	Run Where
Naming Service	CMD
Server	Eclipse
Client	Eclipse

This is the BEST method.

● STEP 1 — Create Project

In Eclipse:

File → New → Java Project

Project name:

DS_CORBA

● STEP 2 — Create Add.idl

Inside src :

New → File

File name:

Add.idl

Paste:

```
module AddModule
{
    interface Add
    {
        long addNumbers(in long a, in long b);
    };
};
```

Save file.

● STEP 3 — Generate CORBA Files

Open CMD:

Go to:

cd C:\Users\HP\eclipse-workspace\DS_CORBA\src

Run:

```
idlj -fall Add.idl
```



What Happens?

CORBA automatically creates:

```
AddModule
```

folder containing:

- stubs
- helper classes
- skeleton classes



STEP 4 — Refresh Eclipse

Right click:

```
DS_CORBA → Refresh
```

Now you should see:

```
AddModule
```

inside `src`.



STEP 5 — Create/Add Java Files

Now paste your uploaded codes:



AddImpl.java

```
import AddModule.*;

public class AddImpl extends AddPOA {

    public int addNumbers(int a, int b) {

        return a + b;
    }
}
```

✓ Server.java

```
import AddModule.*;

import org.omg.CosNaming.*;
import org.omg.CORBA.*;
import org.omg.PortableServer.*;

public class Server {

    public static void main(String args[]) {

        try {

            ORB orb = ORB.init(args, null);

            POA rootpoa =
                POAHelper.narrow(
                    orb.resolve_initial_references(
                        "RootPOA"));

            rootpoa.the_POAManager().activate();

            AddImpl addImpl = new AddImpl();

            org.omg.CORBA.Object ref =
                rootpoa.servant_to_reference(
                    addImpl);

            Add href =
                AddHelper.narrow(ref);

            org.omg.CORBA.Object objRef =
```

```
        orb.resolve_initial_references(  
            "NameService");  
  
        NamingContextExt ncRef =  
            NamingContextExtHelper.narrow(  
                objRef);  
  
        String name = "ADD";  
  
        ncRef.rebind(  
            ncRef.to_name(name),  
            href);  
  
        System.out.println(  
            "CORBA Server Ready...");  
  
        orb.run();  
  
    } catch (Exception e) {  
  
        System.out.println(e);  
    }  
}  
}
```

✓ Client.java

```
import AddModule.*;  
  
import org.omg.CosNaming.*;  
import org.omg.CORBA.*;  
  
import java.util.Scanner;  
  
public class Client {  
  
    public static void main(String args[]) {  
  
        try {  
  
            ORB orb = ORB.init(args, null);  
  
            org.omg.CORBA.Object objRef =  
                orb.resolve_initial_references(  

```

```
        "NameService");

NamingContextExt ncRef =
    NamingContextExtHelper.narrow(
        objRef);

Add addobj =
    AddHelper.narrow(
        ncRef.resolve_str("ADD"));

Scanner sc =
    new Scanner(System.in);

while (true) {

    System.out.print(
        "\nEnter first number: ");

    int a = sc.nextInt();

    System.out.print(
        "Enter second number: ");

    int b = sc.nextInt();

    int result =
        addobj.addNumbers(a, b);

    System.out.println(
        "Result = " + result);

    System.out.print(
        "\nContinue? (yes/no): ");

    String ch = sc.next();

    if (ch.equalsIgnoreCase("no")) {

        break;
    }
}

sc.close();

} catch (Exception e) {

    System.out.println(e);
}
```

}



● STEP 6 — Compile Everything

Open CMD:

```
cd C:\Users\HP\eclipse-workspace\DS_CORBA\src
```

Run:

```
javac *.java AddModule/*.java
```



Expected

No errors.

Warnings are okay.

● STEP 7 — Start Naming Service

Open NEW CMD.

Run:

```
tnameserv -ORBInitialPort 1050
```



KEEP THIS WINDOW OPEN.

● STEP 8 — Run Server from Eclipse

Right click:

Server.java



Run As → Run Configurations

Open:

Arguments

tab.

Inside Program Arguments:

-ORBInitialPort 1050

Click:

Run

☒ Expected Output

CORBA Server Ready...

● STEP 9 — Run Client from Eclipse

Right click:

Client.java



Run As → Run Configurations

Add SAME argument:

-ORBInitialPort 1050

Click:

Run

FINAL OUTPUT

Enter first number: 10

Enter second number: 5

Result = 15

COMPLETE FLOW

Add.idl



idlj generates files



Compile



Start NameService



Run Server



Run Client

